

Inheritance Practice: Vehicle Rental System

Design a Vehicle Rental System for a company that rents out three types of vehicles:

- Car
- Bike
- Truck

All vehicles have the following common information:

- Brand
- Model
- Year of Manufacture
- Rental Price per Day

The rental price per day will be provided when a vehicle object is created.

The company calculates the rental cost differently depending on the type of vehicle.

Rental Rules

Car

- Calculate the rental cost based on the rental price per day and number of rental days.
- If the car is more than 5 years old, apply a 10% discount to the rental cost.

Bike

- Calculate the rental cost based on the rental price per day and number of rental days.
- If the bike is rented for more than 7 days, apply a 15% discount to the total rental cost.

Truck

- Calculate the rental cost based on the rental price per day and number of rental days.
- Add an additional \$100 per day maintenance fee.

OOP Requirements

Design the solution using appropriate OOP concepts.

You should:

1. Create a base Vehicle class containing the common properties and behavior.
2. Create Car, Bike, and Truck classes by inheriting from Vehicle.
3. Define a common method for calculating rental cost.

4. Override the rental-cost calculation method in the derived classes where necessary.
 5. Use constructor chaining so that the constructors of Car, Bike, and Truck initialize the common vehicle information through the Vehicle constructor.
 6. Create several vehicle objects with different rental prices and rental periods.
 7. Calculate and display the final rental cost for each vehicle.
- - Specialization of behavior in derived classes